

```

clc; clear; close all;

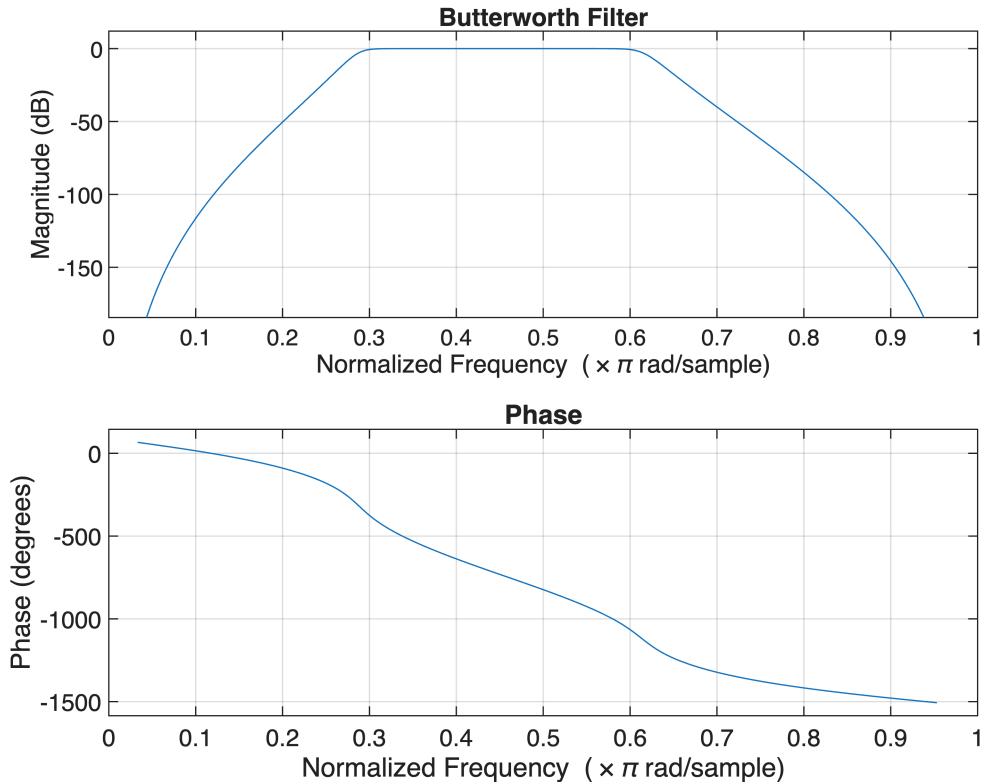
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1, 'bandpass');

[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2, 'bandpass');

figure;
freqz(b1,a1); title('Butterworth Filter');

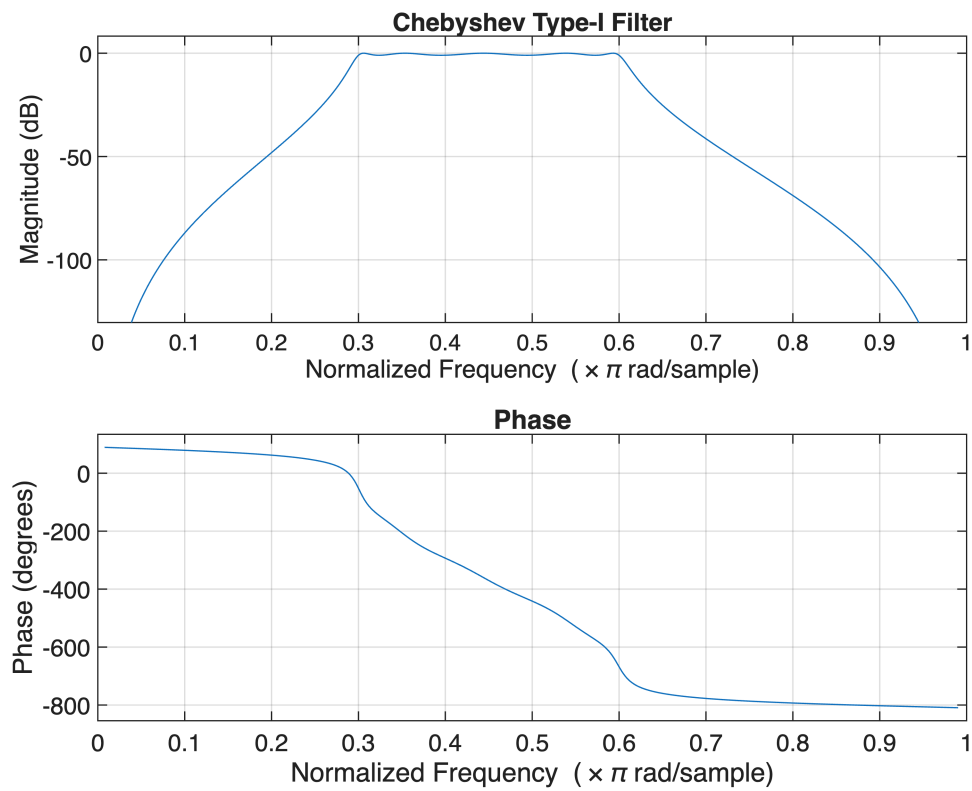
```



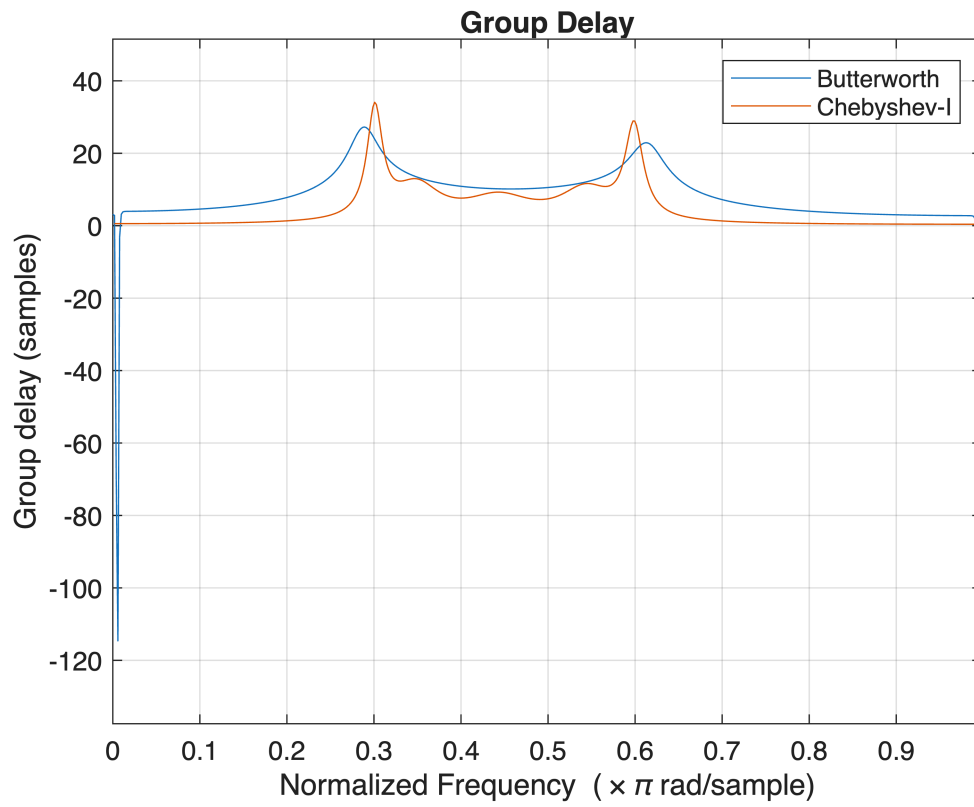
```

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

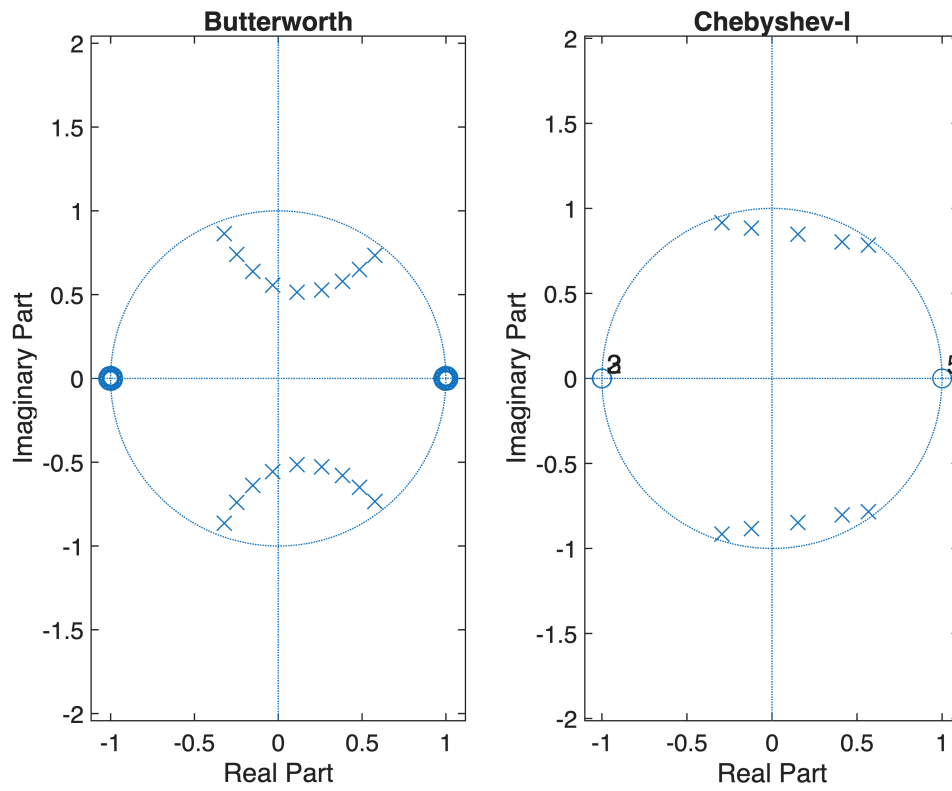
```



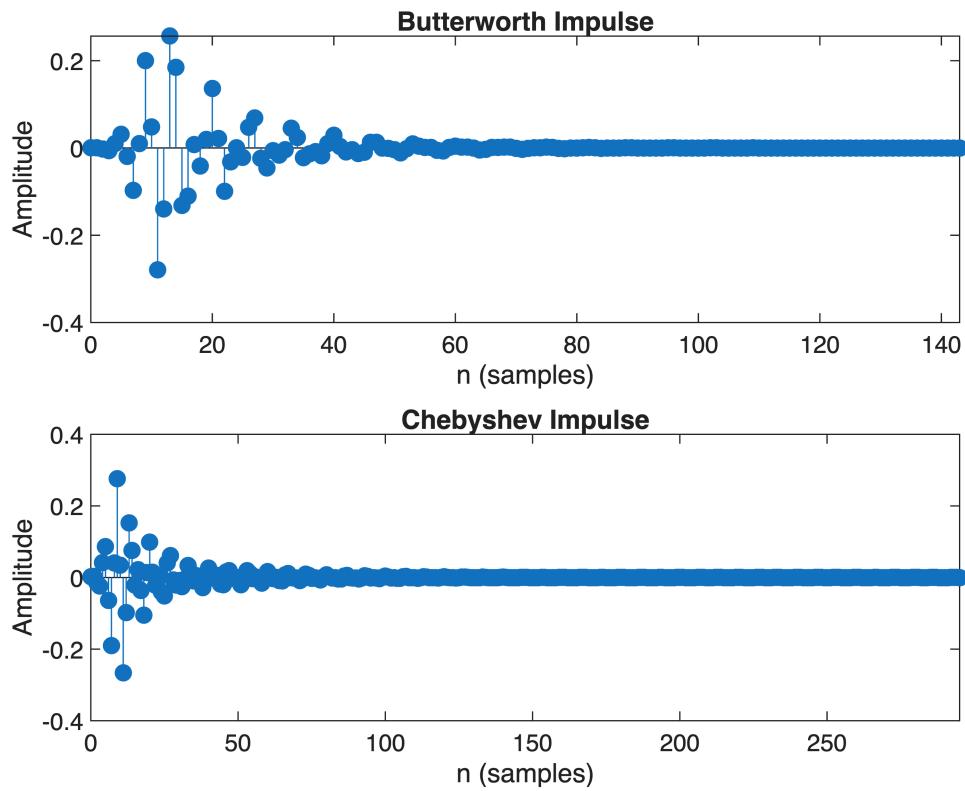
```
figure;  
grpdelay(b1,a1); hold on;  
grpdelay(b2,a2);  
legend('Butterworth','Chebyshev-I');
```



```
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')
```

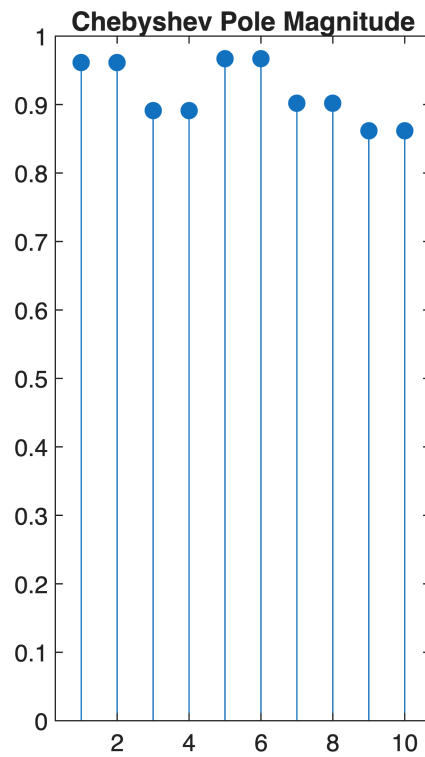
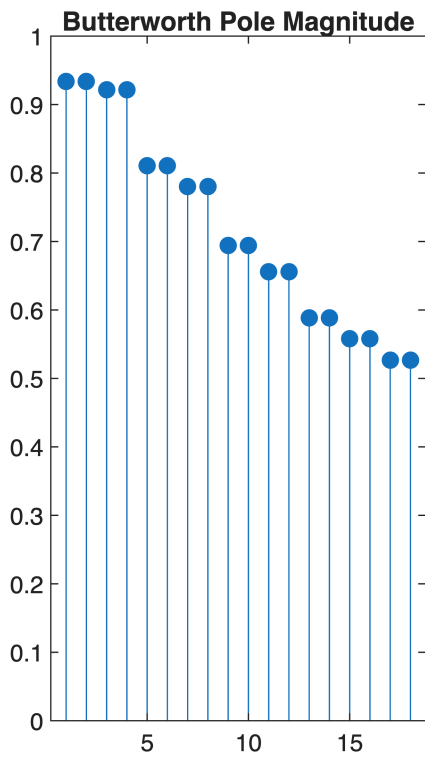


```
figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')
```



```
figure;
subplot(121),stem(abs(roots(a1)),'filled')
title('Butterworth Pole Magnitude')

subplot(122),stem(abs(roots(a2)),'filled')
title('Chebyshev Pole Magnitude')
```



```
disp(['Butterworth Order = ',num2str(n1)])
```

Butterworth Order = 9

```
disp(['Chebyshev Order = ',num2str(n2)])
```

Chebyshev Order = 5